

Notes on Group Homomorphisms¹

After defining groups and beginning to explore their structures, we can organize what we find by constructing a category of groups. In order to do that we need a relation that measures the sameness of groups. Considering a group to be a set with a binary operation, we can begin with a function, the relation that measures the sameness of sets, and add some additional requirement to preserve the properties of the binary operation.

A group homomorphism, then, is a function ϕ acting on a group G such that for all $a, b \in G$, $\phi(ab) = \phi(a)\phi(b)$. That is, the image of a product is equal to the product of images. While it is necessary to understand that the operations on the left- and right-hand sides of this equation may be different, in practice there is little room to confuse them. We further call an injective homomorphism a monomorphism, a surjective homomorphism an epimorphism, a bijective homomorphism an isomorphism, a homomorphism between a group and itself an endomorphism, and an isomorphism between a group and itself an automorphism.

To confirm that a group homomorphism satisfies the conditions of a categorical morphism, we need to check that identity morphisms exist and that the homomorphism property is preserved under composition.

(The Identity Homomorphism) The map that sends everything to itself is a homomorphism and serves as the categorical identity morphism. *Proof.* Let ϕ be the identity map on a group $\mathcal{G}$. For all $a, b \in \mathcal{G}$, $\phi(ab) = ab = \phi(a)\phi(b)$. Understanding how the identity function works, it is trivial to confirm that this acts as the categorical identity.

ἀγάρη

Consider then the morphisms between groups and their subgroups. For groups $\mathcal{G}$ and H , $H < G$ up to isomorphism if and only if there exists a monomorphism $m : H \rightarrow G$. Continuing down this road we might construct and prove a similar theorem for epimorphisms.

(Composition of Homomorphisms) The composition of group homomorphisms is itself a homomorphism. *Proof.* If ϕ and ψ are composable group homomorphisms on $\mathcal{G}$, then for all $ab \in \mathcal{G}$,

$$\phi\psi(ab) = \phi(\psi(ab)) = \phi(\psi(a)\psi(b)) = \phi(\psi(a))\phi(\psi(b)) = \phi\psi(a)\phi\psi(b)$$

χάρά

Next, we wonder what kind of internal structures and properties these maps preserve. We might wonder about the homomorphic image of an inverse or a subgroup. We might wonder if the homomorphic image of an abelian group is abelian or if the homomorphic image of the commutator subgroup is a commutator subgroup. Maybe, the homomorphic

¹Elliott Best, Proofs and Modern Mathematics, *Jackson Hole Classical Academy*, Spring 26.

image of a commutator subgroup is a subset of the commutator subgroup of the image. Or maybe it is something else entirely.

(Image of a Cycle) The homomorphic image of a cyclic subgroup is cyclic. *Proof.* Let $\mathcal{G}$ be a cyclic group generated by g . That is, for all $a \in \mathcal{G}$ there exists an $n \in \mathbb{N}$ such that $g^n = a$. Then for a homomorphism ϕ acting on $\mathcal{G}$, $\phi(a) = \phi(g^n) = \phi(g)^n$. Since every element in the image is generated by the image of the generator, $\phi(\mathcal{G})$ is cyclic. Beyond that, this also proves that the image of a generator is a generator of the image.

είρηνή

(Image of a Subgroup) The homomorphic image of a subgroup is a subgroup. *Proof.* Let H be a subgroup of $\mathcal{G}$, and consider $\phi(H)$. If $x, y \in \phi(H)$, there exist $a, b \in H$ such that $\phi(a) = x$ and $\phi(b) = y$. Then, $xy = \phi(a)\phi(b) = \phi(ab) \in \phi(H)$. Similarly,

$$\phi(e)\phi(a) = \phi(ea) = \phi(a) = \phi(ae) = \phi(a)\phi(e)$$

and

$$\phi(a^{-1})\phi(a) = \phi(a^{-1}a) = \phi(e) = \phi(aa^{-1}) = \phi(a)\phi(a^{-1})$$

The image of the identity is the identity of the image, and the image of an inverse is the inverse of its image. This means we can sensibly write $\phi(a^{-1}) = \phi(a)^{-1}$.

μακροθυμία

For exploration, we might create and prove theorems about abelian groups, centers, and commutator subgroups. At this point, we could also begin to create homomorphisms on groups. However, because we know that groups are built of interrelated cycles, we might wait to start creating them until we understand how they act on cycles a little better. Instead, we will investigate how homomorphisms can relate different parts of a group to the special element of its codomain: the identity.

(The Trivial Homomorphism) The map that sends everything to the identity is a homomorphism. *Proof.* Let ϕ be the map on a group $\mathcal{G}$ that sends everything to the identity. Then for all $a, b \in \mathcal{G}$, $\phi(ab) = e = \phi(a)\phi(b)$.

χρηστότης

The trivial homomorphism can serve as a sort of base case. But what happens when the whole group doesn't map to the identity? The kernel of a group homomorphism, in linear algebra called the null set of a matrix, is the subset of elements that map to the identity.

(The Kernel Subgroup) The kernel of a group homomorphism is a subgroup. *Proof.* If a and b are in the kernel of $\mathcal{G}$ then $\phi(ab) = \phi(a)\phi(b) = e$ and ab is also in the kernel. The identity maps to the identity and is thus in the kernel. And if a is in the kernel, $\phi(a^{-1}) = \phi(a)^{-1} = e^{-1} = e$.

ἀγαθωσύνη

This theorem may be surprisingly easy to prove compared with how strange it may sound. What is this theorem really saying? Actually, quite a bit. This is a first step toward characterizing the homomorphisms that act on a group. We might wonder if every

subgroup is the kernel of some homomorphism. Similarly, we might wonder how many homomorphisms have the same kernel.

What is the kernel of the trivial homomorphism? Because it is constant, the kernel is the whole group. What is the kernel of an isomorphism? Because it is injective, the kernel is trivial. We might wonder if there is a spectrum of the number of homomorphisms for a given kernel. As the kernel shrinks, more elements become available for variation in the homomorphisms. We might wonder how we could characterize these variations.

(Trivial Kernels and Monomorphisms) A homomorphism is a monomorphism if and only if the kernel is trivial. *Proof.* Let ϕ be a homomorphism on $\mathcal{G}$ and suppose that the kernel is the trivial subgroup. If $\phi(a) = \phi(b)$, then $\phi(a)\phi(b)^{-1} = e$ and $\phi(ab^{-1}) = e$. Since the kernel is trivial, $ab^{-1} = e$ and $a = b$. Conversely, suppose that ϕ is injective. Then $\phi(e) = e$ is the unique element of the kernel.

πίστις

(Homomorphic Inverses) A homomorphism is an isomorphism if and only if it has a homomorphic inverse. *Proof.* We know that a function is a bijection if and only if it has an inverse and we assume here that the bijection is a homomorphism. Though this theorem is stated as a biconditional, it suffices to only show the forward conditional because a bijection is its inverse's inverse. Let $\phi : \mathcal{G} \rightarrow H$ be a bijective homomorphism. Then, for all $a, b \in H$ there exists an x, y in $\mathcal{G}$ such that $\phi(x) = a$ and $\phi(y) = b$ and

$$\phi^{-1}(ab) = \phi^{-1}(\phi(x)\phi(y)) = \phi^{-1}(\phi(xy)) = xy = \phi^{-1}(a)\phi^{-1}(b)$$

πραύτης

(Automorphism Groups) The automorphisms on a group form a group under composition. *Proof.* The proof of this theorem is based almost entirely on previous results. Since the compositions of bijections are bijections and the compositions of homomorphisms are homomorphisms, the compositions of automorphisms are automorphisms. The identity map $\iota(x) = x$ is a bijective homomorphism and $\iota\phi(x) = \phi(x) = \phi\iota(x)$. And a map is a bijective homomorphism if and only if it has a bijective, homomorphic inverse.

ἐγκράτεια

(The Inverse Homomorphism and Abelian Groups) A group is abelian if and only if the map that sends each element to its inverse is an automorphism. Notice this means that the automorphism group on an abelian group has an element order 2. What else might we deduce from this simple result? *Proof.* If $\phi : \mathcal{G} \rightarrow \mathcal{G}$, $x \mapsto x^{-1}$ is an automorphism, $ab = \phi(b^{-1}a^{-1}) = \phi(b^{-1})\phi(a^{-1}) = ba$. Conversely, if $\mathcal{G}$ is a group, $\phi : \mathcal{G} \rightarrow \mathcal{G}$, $x \mapsto x^{-1}$ is always a surjection by the existence of inverses and an injection by the uniqueness of inverses. When $\mathcal{G}$ is abelian, $\phi(ab) = b^{-1}a^{-1} = a^{-1}b^{-1} = \phi(a)\phi(b)$.

ζωήν

An interesting way to visualize this inverse map is by reflecting each cycle in the cycle diagram across a central diagonal. We might wonder about other visual representations of more complex automorphisms. One fascinating feature of automorphisms is that there

are some subgroups like the cycles just mentioned that move but always stay within their subgroup. We call a subgroup characteristic if it is equal to its image under every group automorphism.

(Characteristic Subgroups) The center and commutator subgroup are both characteristic. *Proof.* Let $\mathcal{G}$ be a group with center Z and let ϕ be a group automorphism on $\mathcal{G}$. If $a \in Z$, then $\phi(a)\phi(g) = \phi(ag) = \phi(ga) = \phi(g)\phi(a)$ for all $g \in \mathcal{G}$. Therefore, the image of the center is a subset of the center of the image. But since the map is injective every element in the center of the image comes from the center. So, $\phi(Z) = Z$. Similarly, let $\mathcal{G}$ be a group with a commutator subgroup D (the commutator subgroup is often called the derived subgroup) and let ϕ be a group automorphism on $\mathcal{G}$. If a is a commutator in $\mathcal{G}$ where $gha = hg$ for some $g, h \in \mathcal{G}$, then $\phi(g)\phi(h)\phi(a) = \phi(h)\phi(g)$. Then, if homomorphisms preserve commutators, by definition homomorphisms also preserve products of commutators. Therefore, the image of the commutator subgroup is a subset of the commutator subgroup of the image. But, just as the center, the fact that the map is injective ensures that the only elements mapping into the commutator subgroup are coming from the commutator subgroup, so $\phi(D) = D$.

εὐσέβειαν

(Inner Automorphism Subgroup) The set of conjugation maps or inner automorphisms is a subgroup of the automorphism group. *Proof.* Let $\mathcal{G}$ be a group and consider the family of maps $\phi_g(x) = gxg^{-1}$ for all $g \in \mathcal{G}$. For each $x \in \mathcal{G}$ consider the element, $g^{-1}xg$. The image $\phi_g(g^{-1}xg) = g(g^{-1}xg)g^{-1} = x$. Therefore, ϕ_g is surjective. If $\phi_g(a) = \phi_g(b)$, $gag^{-1} = gbg^{-1}$ and $a = b$ by right and left cancelation. Therefore, ϕ_g is injective. For all $a, b \in \mathcal{G}$, $\phi_g(ab) = g(ab)g^{-1} = gag^{-1}gbg^{-1} = \phi_g(a)\phi_g(b)$. Therefore, ϕ_g is an automorphism. Now, if ϕ_g and ϕ_h are both inner automorphisms, then $\phi\psi(x) = \phi(hxh^{-1}) = (gh)x(gh)^{-1}$. The inner automorphism $\phi_e(x) = exe = x$ is the identity. And $\phi_g^{-1}\phi_g(x) = g^{-1}gxg^{-1}g = x$. Thus, the inner automorphisms form a subgroup of the automorphism group.

ὑπομονήν

Notice that the inner automorphism group on an abelian group is trivial. This must then be a metric for exploring the patterns of non-abelian groups. Remember, all we do in math is measure patterns with patterns (with patterns). For those who have eyes to see and ears to hear, what fun.

(Cayley's Theorem) If $\mathcal{G}$ is a group order n , then $\mathcal{G}$ is isomorphic to a subgroup of S_n . *Proof.* For a group $\mathcal{G}$ order n and elements $g \in \mathcal{G}$, consider the family of functions on the underlying sets $f : \mathcal{G} \rightarrow G$ such that $x \mapsto gx$. Each of these functions is represented by a distinct row on the Cayley table of $\mathcal{G}$. Then, by the Sudoku theorem, each function is a bijection on G and thus an element of S_n . Since there is a unique function for each g we can define a map $\phi : \mathcal{G} \rightarrow S_n$ by $g \mapsto f_g$.

$$\phi(ab)(x) = f_{ab}(x) = abx = a(bx) = f_a((f_b)(x)) = \phi(a)\phi(b)x$$

And if $\phi(a) = \phi(b)$, then $f_a = f_b$ and $a = b$. Therefore, ϕ is a monomorphism into S_n indicating that $\mathcal{G}$ is isomorphic to a subgroup. This theorem brings us very close to seeing that symmetry is essential to groups.

φιλαδέλφειαν

(The Alternating Subgroup) The set of even permutations of S_n forms a subgroup called the alternating subgroup, A_n . *Proof.* The number of transpositions in a composition of two even permutations is a sum of even natural numbers which is even. Therefore, the subset is closed. The identity permutation is even. Let $\pi = (a_1 b_1)(a_2 b_2) \dots (a_n b_n)$ be an even permutation. Because all transpositions have order 2 its inverse can be expressed with the same number of transpositions in opposite order, $(a_n b_n)(a_{n-1} b_{n-1}) \dots (a_1 b_1)$. Therefore, the even permutations form a closed subset with identity and inverses.

ἀρετή

Finally, for exploration, we might begin trying to define our own homomorphisms and automorphisms. We've seen the identity automorphism, trivial homomorphism, inverse automorphism, and conjugation automorphisms. Can you invent a few of your own and explore their properties? Maybe there is a way to tweak the bijections in Cayley's theorem to create homomorphisms.

... κατὰ τῶν τοιούτων οὐκ ἔστιν νόμος.